Ram Prasad Bidiyasar

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Experience

Airtel Digital — Gurugram, India
Senior Software Engineer, Xstream backend — *Jul. 2025 – Present*

- Redesigned **Continue Watching** from scratch: watch events keyed by user ID go through Kafka into ScyllaDB, serializing multi-device writes per user, with Redis/Aerospike caches; sync latency fell nearly 70%.
- Re-architected watch-history reads on Spring WebFlux and reactive ScyllaDB (**Watch Info v2**): p99 latency down 75%, per-instance throughput up about 4x; extended it to live TV and microdrama.
- Built the IPTV **multi-CDN routing engine** for 10M+ active users: MongoDB-configured rules over geo, channel, priority, weights and CDN health, with a default-weight fallback, cut buffering from 1% to 0.07%.
- Rebuilt the **event ingestion service** on Java 25: 6x per-pod throughput (∼400 to ∼2,500 req/s) for 80–90K events/s peaks; per-product Kafka clusters and Pub/Sub replay mean no events are lost.
- Designed and built the IPTV **session service** on multi-zone Aerospike: an atomic concurrent-device limit with oldest-device eviction, device-bound tokens and one SDK shared by playback and subscription management; led **device management** on Xstream so users can log out any device.
- Held Home and Rush at **10–11x normal traffic** (India vs New Zealand T20I) with 0% customer-visible errors: an Envoy/Lua gateway filter serves a prepared CDN page after 1.5 s or a 5xx (∼99% availability).
- Scaled page generation from 30 to 100 TPS with nearly 30% lower latency (Java 21, gRPC, parallel async calls); CPU boost and gradual pod warm-up cut autoscaling latency spikes from 900–1000 to 450–550 ms.
- Took over the **UCLM** campaign platform (20+ services, 1.5B+ messages a day): fixed build and load issues and scaled it for the initial migration; resolved Black Duck and Coverity findings in 20+ services.
- Drove **partner onboarding** for 3 partners (Fridaay, Amazon Prime, Hotstar), ingesting 10,944 titles; a reusable integration guide and agent rules cut integration time from 2 weeks to 3 days.
- Built an **AI operations assistant** (Java 25, LangChain4j, Gemini) that investigates missing titles and service errors across MongoDB, ELK and Prometheus, running PROD fixes only after confirmation.

Airtel Digital — Gurugram, India
Software Engineer, Xstream backend — *Oct. 2023 – Jun. 2025*

- Implemented most of the code for the **AWS to GCP migration** of 20+ Xstream services (SQS/Kinesis to Kafka, S3 to GCS, MongoDB, ScyllaDB), with load-tested Kafka partitions and no downtime.
- Built a **Debezium and Kafka** pipeline from the Xstream catalog to data science, cutting enrichment delay from 1–2 days to near real time, plus event-driven episode promotion for watched series.
- Upgraded package and discovery services to Java 21 and Spring Boot 3.5; traced hourly 2–3 minute user-service p99 spikes to log-file rollover.
- Adapted Prebid Server (Vert.x) with Druid analytics; built a Keycloak role-based onboarding module.

Technical Skills

Languages and frameworks: Java, C++, SQL, Spring Boot, Spring WebFlux, gRPC, Vert.x, Kafka Streams
Data and messaging: Kafka, Kafka Connect, ScyllaDB, Aerospike, PostgreSQL, MongoDB, Redis, Elasticsearch, Druid, Pub/Sub
Cloud and infra: GCP, AWS, Kubernetes, Docker, Envoy, Jenkins, Airflow
Practices: distributed systems, microservices, failover design, observability (Grafana, Kibana, AppDynamics)

Education

Malaviya National Institute of Technology Jaipur — Jaipur, India
B.Tech, Electronics and Communication Engineering (CGPA 8.16) — *Jul. 2019 – May 2023*